Problem 1

a) $f(x)=\sum_{j=1}^{p}\theta_j h_j(x)$ $h_j(x)=\max(0,x-b_j)$ $\theta_j \overset{iid}{\sim} N(0,T^2)$

$f(x)=h(x)^T\theta$, $h(x)=(h_1(x)\dots h_p(x))^T$, $\theta=(\theta_1\dots\theta_p)^T\sim N(0,T^2 I_p)$

$(f(x_1)\dots f(x_m))^T=H\theta$, $H(m\times p)$ w/ $H_{ij}=h_j(x_i)$

Since θ MVN $\rightarrow (f(x_1)\dots f(x_m))^T$ MVN

$\therefore \{f(x)\}_{x\in[0,1]}$ is a Gaussian Process

Mean: $E[f(x)]=E[\sum_{j=1}^{p}\theta_j h_j(x)]=\sum h_j(x)E[\theta_j]=0$

Cov: $\mathrm{Cov}(\theta_j,\theta_k)=\begin{cases}T^2, & j=k\\ 0 & j\neq k\end{cases}$, independent & uncoorelated

$\mathrm{Cov}(f(x),f(x'))=\mathrm{cov}(\sum_{j=1}^{p}\theta_j h_j(x),\sum_{k=1}^{p}\theta_k h_k(x'))$

$=\sum_{j=1}^{p}T^2 h_j(x)h_j(x')=T^2\sum h_j(x)h_j(x')=T^2\langle h_j(x),h_j(x')\rangle$

Kernel is $K(x,x')=\langle h(x),h(x')\rangle=\sum_{j=1}^{p}h_j(x)h_j(x')$

$\therefore \mathrm{Cov}(f(x),f(x'))=T^2K(x,x')$

f is a zero-mean GP w/ cov kernel $T^2K(x,x')$

b) $k_0=\begin{pmatrix}K(x_0,x_1)\\ \vdots\\ K(x_0,x_n)\end{pmatrix}$, $y=f(x_0)$, $\begin{pmatrix}Y\\ y_0\end{pmatrix}\sim N\left(0,\begin{pmatrix}T^2K+\sigma^2 I & T^2k_0\\ T^2k_0^T & T^2k_{00}\end{pmatrix}\right)$

Using the conditional gaussian formula

$y_0|x_0,D\sim N(\mu_0(x_0),\sigma^2(x_0))$

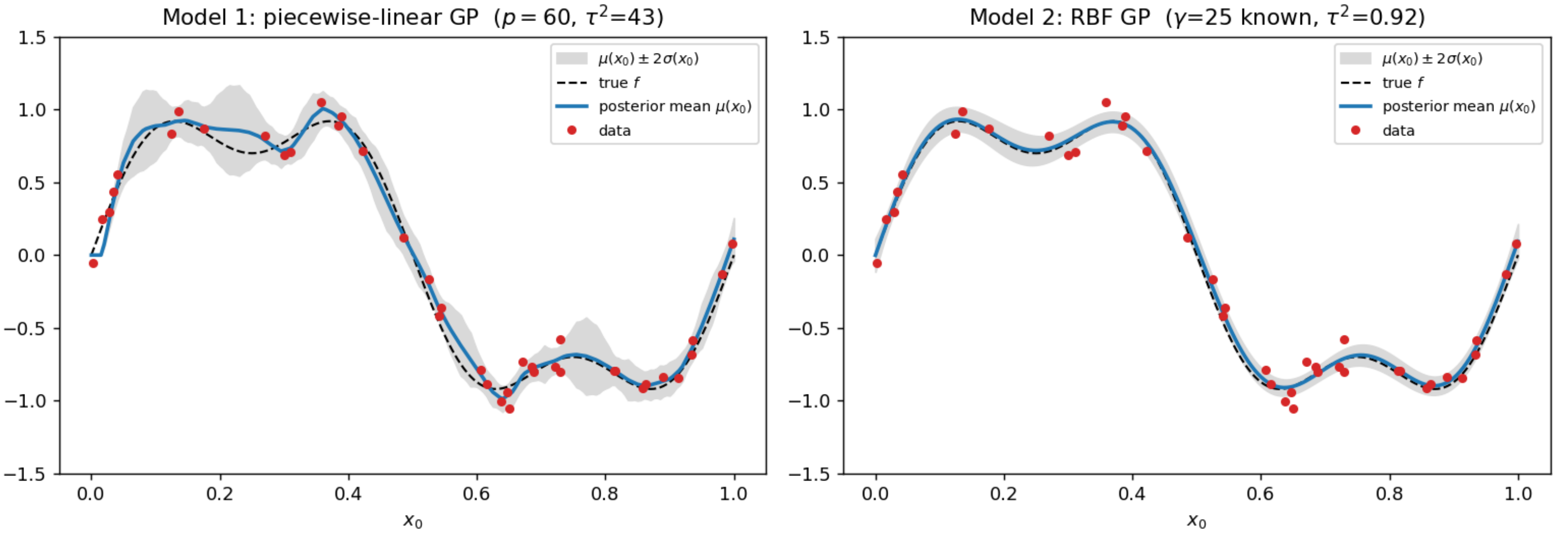
$$\boxed{\begin{aligned}\mu(x_0)&=T^2k_0^T(T^2K+\sigma^2 I_n)^{-1}Y\\ \sigma^2(x_0)&=T^2k_{00}-T^4k_0^T(T^2K+\sigma^2 I_n)^{-1}k_0\end{aligned}}$$

c) $\boxed{T^2=42.9}$

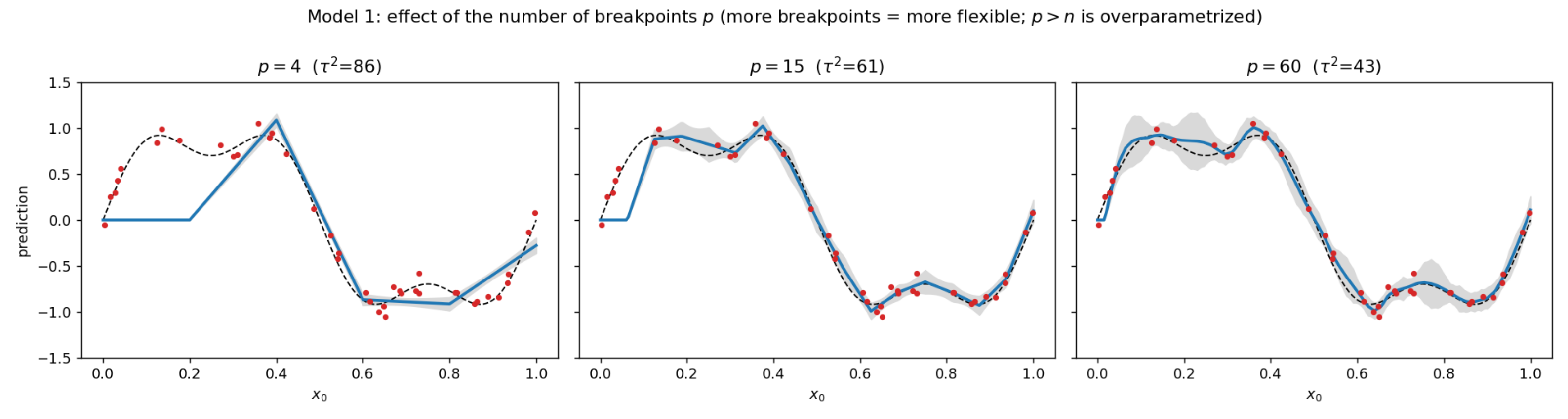
As p increases the model becomes more flexible, small p less adaptive, while large p captures more shape, even when $p>n$ the fit stays controlled because T^2 is re-selected by the evidence which regularizes the size of the slope-change coeff

For the band:

Narrower near observed data points, especially where many x_i close together. Wider near gaps between data points where there is less information



Model 1: piecewise-linear GP ($p = 60$, τ^2=43)
Model 2: RBF GP (γ=25 known, τ^2=0.92)
$\mu(x_0) \pm 2\sigma(x_0)$
true f
posterior mean $\mu(x_0)$
data
x_0


Model 1: effect of the number of breakpoints p (more breakpoints = more flexible; $p > n$ is overparametrized)
$p = 4$ (τ^2=86)
$p = 15$ (τ^2=61)
$p = 60$ (τ^2=43)
prediction
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0.0
0.2
0.4
0.6
0.8
1.0
x_0